INCH-POUND

MIL-A-70604 (AR)
AMENDMENT 2
01 March 1996
SUPERSEDING
AMENDMENT 1
23 November 1990

MILITARY SPECIFICATION

AMINE BONDING AGENT

This Amendment forms a part of Military Specification MIL-A-70604 (AR), dated 22 April 88, and is approved for use by the US Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.1 Scope. Change "one type" to "two types".

Add the following paragraph:

* "1.2 Classification. The amine bonding agent shall be of the following types:

Type I - For use in the M864 system.
Type II - For use in the M913 system."

* 2.1.1 Specifications, standards and handbooks. Delete:

"MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes"

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* 2.2 Other Publications. Add the following:

"ANSI/ASQC Z1.4-1003 - Sampling Procedure and Tables for inspection by Attributes

(Application for copies should be addressed to: American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036).

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* 3.1 Composition. Delete in its entirety and substitute the following:

* "3.1 Composition. Type I amine bonding agent is the result of the reaction between tetraethylenepentamine and acrylonitrile to form Tepan. Type II amine bonding agent is the result of reaction of Tepan with glycidol to form Tepanol. The type I or II material shall conform to the requirements of Table I."

Change Table I to the following:

"TABLE I. Chemical and Physical Properties for Amine Bonding Agents

	Requirement				
	Type I		Type II		Applicable
Property	Min	Max	Min	Max	Test Paragraph
Total Amine, eq/100g	1.1	1.6	0.8	1.2	4.5.2
Ammonia, ppm	---	500	---	500	4.5.4
Moisture, %	---	0.85	---	0.5	4.5.5
Viscosity, 25 C, cps	500	1,000	30,000	60,000	4.5.6
Hydroxyl Content, eq/100g"	NA	NA	0.4	0.6	4.5.9

* 3.2 Infrared Spectrum. Delete in its entirety and substitute the following:

"3.2 Infrared Spectrum. The IR spectrum of the sample for the Type I amine bonding agent shall contain the major peaks shown in Figure I when tested in accordance with 4.5.7."

Change the title of Figure 1 from "Figure 1 - IR Spectrum for Amine Bonding Agent" to "Figure 1 - IR Spectrum for Type I Amine Bonding Agent"

* 4.3.2 Inspection to be performed. Delete Table II in its entirety and substitute the new Table II included.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER / NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Total Amine		3.1	4.5.2
	Ammonia		3.1	4.5.4
	Moisture		3.1	4.5.5
	Viscosity		3.1	4.5.6
	Infrared spectrum, (Type I only)		3.2	4.5.7
	Hydroxly Content, (Type II only)		3.1	4.5.9

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* 4.4.1 Lot formation. Delete “MIL-STD-105” and replace with “ANSI/ASQC Z1.4”

4.4.3.1 Sampling. Change sample size table to the following:

"Number of Containers in Batch	Number of Containers to be Sampled
2 - 200	2
Over	5"

* Delete Table III in its entirety and substitute the following:

"TABLE III. Classification of Defects.

Test/Examination	Requirement Paragraph	Defect Classification
Total amine	3.1	Major
Ammonia	3.1	Major
Moisture	3.1	Major
Viscosity	3.1	Major
Infrared spectrum (Type I Only)	3.2	Major
Hydroxyl content (Type II Only)	3.1	Major
Workmanship	3.4	Major"

4.5.1 Secondary amine. Delete in its entirety.

4.5.3 Tertiary amine. Delete in its entirety.

4.5.4.2 a. Delete in its entirety and substitute the following:

" a. Sodium hydroxide, 10M. Add 40 grams of AR Grade NaOH to 80 ml of DI water in a 100 ml volumetric flask, dissolve, and dilute to volume with deionized water."

Add the following paragraphs:

* "4.5.9 Hydroxyl content. The hydroxyl content for amine bonding agent (Type II only) shall be determined as follows:

4.5.9.1 Equipment.

a. Steam bath.

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b. Flask, 250 ml with ground glass joints and stoppers.

c. Buret, 50 ml.

d. Pipette, 20 ml.

e. Balance, analytical.

4.5.9.2 Reagents.

a. Acetic Anhydride, AR grade.

b. Pyridine, AR grade.

c. Sodium Hydroxide, AR grade.

d. Acetylating reagent. Add 30 ml of acetic acid to 470 ml pyridine, mix thoroughly. Prepare fresh daily. Store in a dark bottle.

e. Sodium Hydroxide solution 0.5 N in methanol, standardized.

f. Phenolphthalein indicator, 1 % in alcohol.

g. DI water.

4.5.9.3 Procedure.

NOTE: Run sample in triplicate. Dry all glassware in 110 C oven prior to use.

a. Weigh a 0.4 to 0.5 gm sample of Type II amine bonding agent to the nearest 0.0001 gm into a 250 ml flask. Record the weight as Ws.

b. Using a 20 ml volumetric pipette, add exactly 20 ml of acetylating reagent to each flask.

c. Prepare three blanks containing 20 ml aliquot of acetylating reagent and carry blanks through the entire treatment of the sample.

d. Moisten glass stoppers with pyridine and seat loosely.

e. Assure sample is completely dissolved in acetylating reagent.

f. Heat samples and blanks gently on steam bath for one hour.

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g. Add 50 ml DI water, rinsing stopper and inside walls of flask.

h. Swirl the contents to ensure thorough mixing and heat on steam bath for five (5) minutes.

i. Remove flasks and cool to room temperature.

j. Rinse stoppers and inside walls with 5-10 ml DI water.

k. Add one-half dropper of phenolphthalein indicator and titrate with standard 0.5N sodium hydroxide to the appearance of the first permanent pink coloration. Record the volume of the blanks, Vb, and samples, Vs.

NOTE: Volume of the blanks must agree within 0.1 ml.

l. Calculate the hydroxyl content as follows:

$$\text{Hydroxyl Content,} \quad \text{eq} / 100\text{g} = \frac{(V_b - V_s) \times N}{10 \times W_s} - N_h$$

Where:

V_b = the average volume of the titrant, sodium hydroxide for the blanks.
V_s = the average volume of titrant, sodium hydroxide for the sample.
N = normality of the titrant, sodium hydroxide.
W_s = weight of the sample.
N_h = secondary amine content, eq/100g."

* 5.2 Marking. Change "LONG TERM STORAGE SHOULD BE BELOW 32 F TO PRESERVE PRODUCT QUALITY" to the following: "LONG TERM STORAGE SHOULD BE AT OR BELOW 0 F TO PRESERVE PRODUCT QUALITY."

* 6.1 Intended use. Delete in its entirety and substitute the following:

"6.1 Intended use. Type I amine bonding agent is intended for use in the base-bleed unit of the M864. Type II amine bonding agent is intended for use in the M913 rocket motor."

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